Yarim va chorak doira perimetri

2-test (qiyinroq) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

---

1. **Yarim doiraning radiusi 2,5 dm. Uning perimetrini toping (π = 3,14).**

A) 7,85 dm **B)** 12,85 dm **C)** 20,7 dm **D)** 10,35 dm

---

2. **Doiraning diametri 4,2 m. Shu doiraning choragi perimetrini toping (π = $\frac{22}{7}$).**

A) 10,8 m **B)** 7,5 m **C)** 15 m **D)** 5,4 m

---

3. **Aylananing $\frac{3}{4}$ qismidan iborat shaklning radiusi 6,3 cm. Shaklning perimetrini toping (π = $\frac{22}{7}$).**

A) 32,4 cm **B)** 42,3 cm **C)** 36 cm **D)** 29,7 cm

---

4. **Chorak doiraning radiusi 8,5 cm. Uning perimetrini toping (π = 3,141). Javobni yuzdan birlar xonasigacha yaxlitlang.**

A) 30,35 cm **B)** 30,34 cm **C)** 13,35 cm **D)** 21,85 cm

---

5. **Shakl ikkita yarim doiradan iborat. Ularning diametrlari 8 cm va 6 cm, ikkalasi bitta to'g'ri chiziqda yonma-yon yotadi. Shaklning perimetrini toping (π = 3,14).**

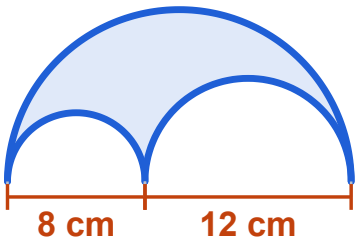
A) 43,96 cm **B)** 35,98 cm **C)** 21,98 cm **D)** 29,98 cm

---

6. **Bo'yalgan shakl bitta katta va ikkita kichik yarim aylana bilan chegaralangan. Uning perimetrini toping (π = 3,14).**

A) 43,96 cm **B)** 31,4 cm

C) 62,8 cm **D)** 82,8 cm



---

7. **Kvadratning tomoni 11 cm. Uning bir uchini markaz qilib, radiusi 11 cm bo'lgan chorak doira kesib olindi. Qolgan shaklning perimetrini toping (π = 3,14).**

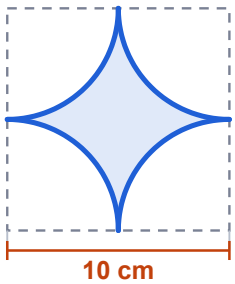
A) 50,27 cm **B)** 61,27 cm **C)** 44 cm **D)** 39,27 cm

---

8. **Kvadratning tomoni 10 cm. Uning har bir uchini markaz qilib, radiusi 5 cm bo'lgan chorak aylana chizildi. Bo'yalgan shaklning perimetrini toping (π = 3,14).**

A) 71,4 cm **B)** 7,85 cm

C) 31,4 cm **D)** 62,8 cm



---

9. **Ayvon yarim doira shaklida, diametri 4,2 m. Diametr tomoni uyning devoriga tegib turadi. Ayvonning qolgan cheti bo'ylab panjara qilinadi. Panjara uzunligi qancha ($\pi = \frac{22}{7}$)?**

A) 13,2 m **B)** 10,8 m **C)** 6,6 m **D)** 8,7 m

---

10. **Chorak doira shaklidagi stol ustining cheti bo'ylab lenta yopishtiriladi. Stolning radiusi 9 dm. Lentaning 1 dm i 1 500 so'm. Lenta uchun necha so'm kerak (π = 3,14)?**

A) 48 195 so'm **B)** 34 695 so'm

C) 21 195 so'm **D)** 69 390 so'm

---

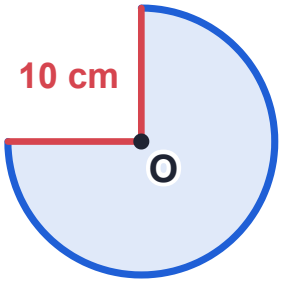
11. **Yarim doiraning diametri 20 cm, chorak doiraning radiusi 20 cm. Qaysi birining perimetri katta va qanchaga (π = 3,14)?**

A) Yarim doiraniki, 20 cm ga **B)** Chorak doiraniki, 20 cm ga

C) Ikkalasi teng **D)** Chorak doiraniki, 31,4 cm ga

---

12. **Rasmda aylananing $\frac{3}{4}$ qismi, O — markaz. Shaklning perimetrini toping (π = 3,14).**

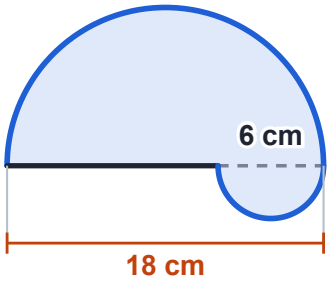


A) 67,1 cm **B)** 57,1 cm

C) 47,1 cm **D)** 82,8 cm

---

13. **Katta yarim doiraning diametri 18 cm. Diametrning o'ng uchidagi 6 cm li qismiga pastdan kichik yarim doira yopishtirildi. Shaklning perimetrini toping (π = 3,14).**



A) 55,68 cm **B)** 37,68 cm

C) 43,68 cm **D)** 49,68 cm

---

14. **Yarim doiraning diametri 7 m. Uning perimetrini π orqali ifodalang.**

A) (3,5π + 3,5) m **B)** (3,5π + 7) m

C) 3,5π m **D)** (7π + 7) m

---

15. **Radiusi 4 cm bo'lgan ikkita bir xil chorak doira radiuslari bilan yonma-yon qo'yilib, yarim doira hosil qilindi. Ikkala chorak doira perimetrlarining yig'indisi hosil bo'lgan yarim doiraning perimetridan qancha katta (π = 3,14)?**

A) 4 cm **B)** 8 cm **C)** 12,56 cm **D)** 0 cm

---

16. **Aylananing $\frac{3}{4}$ qismidan iborat shaklning radiusi 5 cm. Perimetrini kalkulyatordagi π tugmasi bilan toping va yuzdan birlar xonasigacha yaxlitlang.**

A) 33,55 cm **B)** 23,56 cm **C)** 33,56 cm **D)** 33,6 cm

---

17. **Usta simdan radiusi 2,1 m bo'lgan yarim doira shaklidagi 5 ta bir xil ramka yasadi. Har bir ramka yarim doira chegarasi bo'ylab egiladi. Jami necha metr sim ketdi ($\pi = \frac{22}{7}$)?**

A) 33 m **B)** 27 m **C)** 54 m **D)** 75 m

---

18. **Ikkita bir xil yarim doiraning diametrlari 10 cm, ular bitta to'g'ri chiziqda ketma-ket yotadi: biri chiziqdan yuqorida, ikkinchisi pastda. Hosil bo'lgan shaklning perimetrini π orqali ifodalang.**

A) (10π + 20) cm **B)** (10π + 10) cm

C) 10π cm **D)** (20π + 20) cm

---

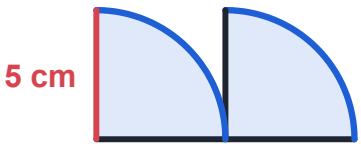
19. **Lola radiusi 8 cm bo'lgan chorak doiraning perimetrini shunday topdi: $\frac{1}{4} \cdot 2 \cdot 3{,}14 \cdot 8 + 16 = 28{,}56$ cm. Uning javobi haqida nima deyish mumkin?**

A) Noto'g'ri, 36,56 cm **B)** Noto'g'ri, 20,56 cm

C) Noto'g'ri, 12,56 cm **D)** To'g'ri: 16 cm — ikkita radius

---

20. **Shakl ikkita bir xil chorak doiradan tuzilgan (rasm). Ularning radiusi 5 cm. Shaklning perimetrini toping (π = 3,14).**



A) 15,7 cm **B)** 25,7 cm

C) 35,7 cm **D)** 30,7 cm

Javoblar

Yarim va chorak doira perimetri · 2-test (qiyinroq)

№	Javob	Qiymati
1	**B**	12,85 dm
2	**B**	7,5 m
3	**B**	42,3 cm
4	**A**	30,35 cm
5	**B**	35,98 cm
6	**C**	62,8 cm
7	**D**	39,27 cm
8	**C**	31,4 cm
9	**C**	6,6 m
10	**A**	48 195 so'm

№	Javob	Qiymati
11	**B**	Chorak doiraniki, 20 cm ga
12	**A**	67,1 cm
13	**D**	49,68 cm
14	**B**	(3,5π + 7) m
15	**B**	8 cm
16	**C**	33,56 cm
17	**C**	54 m
18	**A**	(10π + 20) cm
19	**D**	To'g'ri: 16 cm — ikkita radius
20	**C**	35,7 cm

Qisqa yechimlar

1. Diametr: 2 · 2,5 = 5 dm.
Yoy: $\frac{1}{2}$ · 3,14 · 5 = 7,85 dm.
Perimetr: 7,85 + 5 = **12,85 dm**.
(10,35 — diametr o'rniga radius qo'shilgan.)

---

2. Radius: 4,2 : 2 = 2,1 m.
Yoy: $\frac{1}{4}$ · 2 · $\frac{22}{7}$ · 2,1 = 3,3 m.
Perimetr: 3,3 + 2,1 + 2,1 = **7,5 m**.
(15 — diametr radius deb olingan.)

---

3. Yoy: $\frac{3}{4}$ · 2 · $\frac{22}{7}$ · 6,3 = 29,7 cm.
Ikkita radius: 29,7 + 6,3 + 6,3 = **42,3 cm**.
(36 — bitta radius qo'shilgan.)

---

4. Yoy: $\frac{1}{4}$ · 2 · 3,141 · 8,5 = 13,34925 cm.
Perimetr: 13,34925 + 8,5 + 8,5 = 30,34925 cm.
Yuzdan birlar xonasidan keyingi raqam 9, shuning uchun: **30,35 cm**.
(30,34 — raqamlar shunchaki tashlab yuborilgan.)

---

5. Yoylar: $\frac{1}{2}$ · 3,14 · 8 = 12,56 cm va $\frac{1}{2}$ · 3,14 · 6 = 9,42 cm.
Pastdagi to'g'ri chiziq: 8 + 6 = 14 cm.
Perimetr: 12,56 + 9,42 + 14 = **35,98 cm**.

---

6. Katta yarim aylananing diametri: 8 + 12 = 20 cm, yoyi: $\frac{1}{2}$ · 3,14 · 20 = 31,4 cm.
Kichik yoylar: $\frac{1}{2}$ · 3,14 · 8 = 12,56 cm va $\frac{1}{2}$ · 3,14 · 12 = 18,84 cm.
To'g'ri kesma yo'q: 31,4 + 12,56 + 18,84 = **62,8 cm**.

---

7. Yoy: $\frac{1}{4}$ · 2 · 3,14 · 11 = 17,27 cm. U ichkariga botgan bo'lsa ham, uzunligi shu.
Kesilgan uchga tutashgan ikki tomon butunlay ketdi, kvadratning qolgan 2 tomoni: 11 + 11 = 22 cm.
Perimetr: 17,27 + 22 = **39,27 cm**.
(61,27 — kvadratning 4 tomoni ham qo'shilgan.)

---

8. Har bir yoy: $\frac{1}{4} \cdot 2 \cdot 3{,}14 \cdot 5 = 7{,}85$ cm.
Shaklni faqat 4 ta yoy o'rab turadi: $4 \cdot 7{,}85 =$ **31,4 cm**.
(4 ta chorak birga — radiusi 5 cm bo'lgan butun aylana. 71,4 — kvadrat tomonlari ham qo'shilgan.)

---

9. Devor tomonida (diametrda) panjara kerak emas — faqat yoy bo'ylab.
$\frac{1}{2} \cdot \frac{22}{7} \cdot 4{,}2 =$ **6,6 m**.
(10,8 — diametr ham qo'shilgan, bu yerda u kerak emas.)

---

10. Yoy: $\frac{1}{4} \cdot 2 \cdot 3{,}14 \cdot 9 = 14{,}13$ dm.
Perimetr: $14{,}13 + 9 + 9 = 32{,}13$ dm.
Narxi: $32{,}13 \cdot 1\,500 =$ **48 195 so'm**.

---

11. Yarim doira: $\frac{1}{2} \cdot 3{,}14 \cdot 20 + 20 = 31{,}4 + 20 = 51{,}4$ cm.
Chorak doira: $\frac{1}{4} \cdot 2 \cdot 3{,}14 \cdot 20 + 20 + 20 = 31{,}4 + 40 = 71{,}4$ cm.
Yoylar teng, farq to'g'ri chiziqlarda: **chorak doiraniki 20 cm ga katta**.

---

12. Yoy: $\frac{3}{4} \cdot 2 \cdot 3{,}14 \cdot 10 = 47{,}1$ cm.
Ikkita radius: $47{,}1 + 10 + 10 =$ **67,1 cm**.

---

13. Katta yoy: $\frac{1}{2} \cdot 3{,}14 \cdot 18 = 28{,}26$ cm. Kichik yoy: $\frac{1}{2} \cdot 3{,}14 \cdot 6 = 9{,}42$ cm.
Katta diametrning 6 cm li qismi shakl ichida (punktir), chegarada qolgani: $18 - 6 = 12$ cm.
Perimetr: $28{,}26 + 9{,}42 + 12 =$ **49,68 cm**.
(55,68 — butun diametr qo'shilgan.)

---

14. Yoy: $\frac{1}{2} \cdot \pi \cdot 7 = 3{,}5\pi$ m.
Diametrni qo'shamiz: **(3,5π + 7) m**.

---

15. Ikkala chorak: $2 \cdot (6{,}28 + 4 + 4) = 28{,}56$ cm.
Yarim doira: $\frac{1}{2} \cdot 3{,}14 \cdot 8 + 8 = 20{,}56$ cm.
Farq: $28{,}56 - 20{,}56 =$ **8 cm** — tutashgan radius ikki marta sanalgan edi ($2 \cdot 4$).

---

16. Yoy: $\frac{3}{4} \cdot 2 \cdot \pi \cdot 5 \approx 23{,}5619$ cm.
Perimetr: $23{,}5619 + 5 + 5 \approx 33{,}5619$ cm ≈ **33,56 cm**.
(33,55 — π = 3,14 bilan chiqadi.)

---

17. Diametr: $2 \cdot 2{,}1 = 4{,}2$ m. Yoy: $\frac{1}{2} \cdot \frac{22}{7} \cdot 4{,}2 = 6{,}6$ m.
Bitta ramka: $6{,}6 + 4{,}2 = 10{,}8$ m.
5 ta: $10{,}8 \cdot 5 =$ **54 m**.

---

18. Har bir yoy: $\frac{1}{2} \cdot \pi \cdot 10 = 5\pi$ cm, ikkitasi 10π cm.
Yarim doiralar bir-birini yopmaydi, ikkala diametr ham chegarada: $10 + 10 = 20$ cm.
Perimetr: **(10π + 20) cm**.

---

19. Yoy: $\frac{1}{4} \cdot 2 \cdot 3{,}14 \cdot 8 = 12{,}56$ cm.
Ikkita radius: $8 + 8 = 16$ cm.
$12{,}56 + 16 = 28{,}56$ cm — **javob to'g'ri**.

---

20. Har bir yoy: $\frac{1}{4} \cdot 2 \cdot 3{,}14 \cdot 5 = 7{,}85$ cm, ikkitasi 15,7 cm.
To'g'ri chiziqlar: pastda $5 + 5 = 10$ cm, chapda 5 cm va o'rtadagi tik radius 5 cm — uning chap tomonida shakl yo'q, demak u ham chegarada.
Perimetr: $15{,}7 + 10 + 5 + 5 =$ **35,7 cm**.
(30,7 — o'rtadagi radius ichki chiziq deb tashlab ketilgan.)
